AIML231 —Techniques in Machine Learning

Week 3

Classification: Performance Metrics

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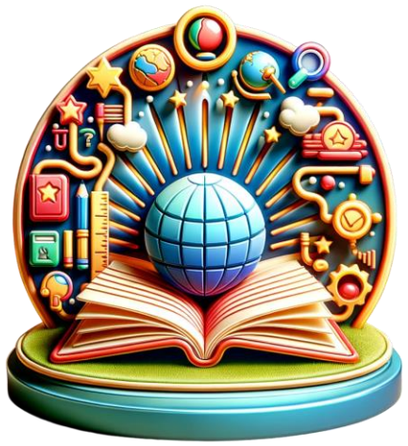
Overview

★ Basic Performance Evaluation Metrics

- Accuracy and Error Rate
- True Positive Rate (TPR) / Sensitivity, True Negative Rate (TNR) / Specificity, False Positive Rate (FPR), False Negative Rate (FNR)
- Precision, Recall and F1-Score
- ROC and AUC

★ Decision Tree



Basic Performance Evaluation Metrics

Classification Accuracy

- Accuracy –the proportion of correctly classified objects relative to the total number of objects in the dataset, expressed as a percentage

$$Accuracy = \frac{N_{CorrClassified}}{N_{total}} * 100\%$$

  - $N_{CorrClassified}$: the number of objects correctly detected or classified
  - N_{total}: the total number of objects in the data set

- Error Rate: the proportion of incorrectly classified objects as a percentage of the number of objects

Confusion Matrix

- A confusion matrix is a table that summarizes the performance of a classifier by comparing predicted labels with actual labels
  - a typical structure like:

Predicted / Actual	Actual: Positive	Actual: Negative
Predicted: Positive	True Positive (TP)	False Positive (FP)
Predicted: Negative	False Negative (FN)	True Negative (TN)

  - allow for a detailed breakdown of errors rather than relying solely on accuracy
  - provide insight into the types of mistakes model makes
  - help compute key performance metric

TPR, TNR, FPR and FNR

- Example - Confusion Matrix on disease diagnose

		Actual Situation	
		Disease	Non-Disease
Diagnosed Situation	Disease	True *Positive* (**TP**)	False Positive (**FP**)
	Non-Disease	False Negative (**FN**)	True Negative (**TN**)

- TPR – True Positive Rate (Fraction) – **sensitivity**
  - The fraction of positive instances (actual Disease) that are correctly classified/detected by a classifier/detector
  - TPR = *TP/(TP+FN)*
- TNR – True Negative Rate (Fraction) – **specificity**
  - The fraction of negative instance (actual non-Disease) that are correctly classified/detected
  - TNR = *TN/(FP+TN)*

TPR, TNR, FPR and FNR

- FPR – False Positive Rate (Fraction)
  - The fraction of negative instances that are incorrectly classified
  - FPR = *FP/(FP+TN)*
- FNR – False Negative Rate (Fraction)
  - The fraction of positive instances that are incorrectly classified
  - FNR = *FN/(TP+FN)*
- FNR + TPR = 1, FPR + TNR = 1, so either (TPR and TNR) or (FNR and FPR) are employed
- The bigger the TPR and TNR, the better the classifier.

True Class

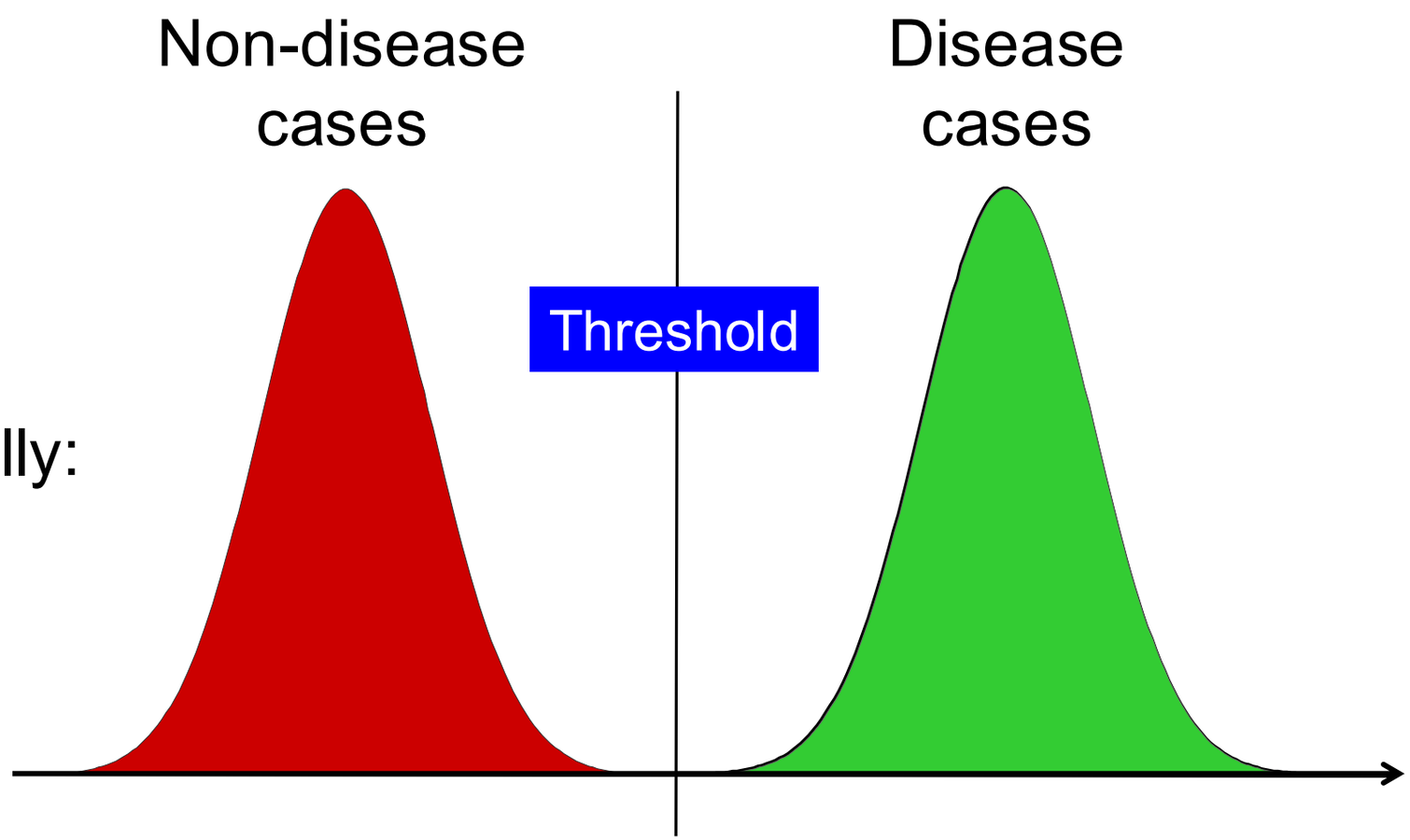
Predicted Class	A	B	C
A	TP_A	E_{BA}	E_{CA}
B	E_{AB}	TP_B	E_{CB}
C	E_{AC}	E_{BC}	TP_C

Example Question

- Suppose a classifier is applied to a two-class object classification problem: class1 (positive) and class2 (negative).
- 200 objects for class1 and 300 objects for class2
- With some threshold, the classifier correctly classified 160 objects for class1 and 210 objects for class2 .
  - What is the accuracy?
  - What is the error rate?
  - What is the TPR?
  - What is the FPR?
  - What is the TNR?
  - What is the FNR?

Receiver Operating Characteristic (ROC) Curve

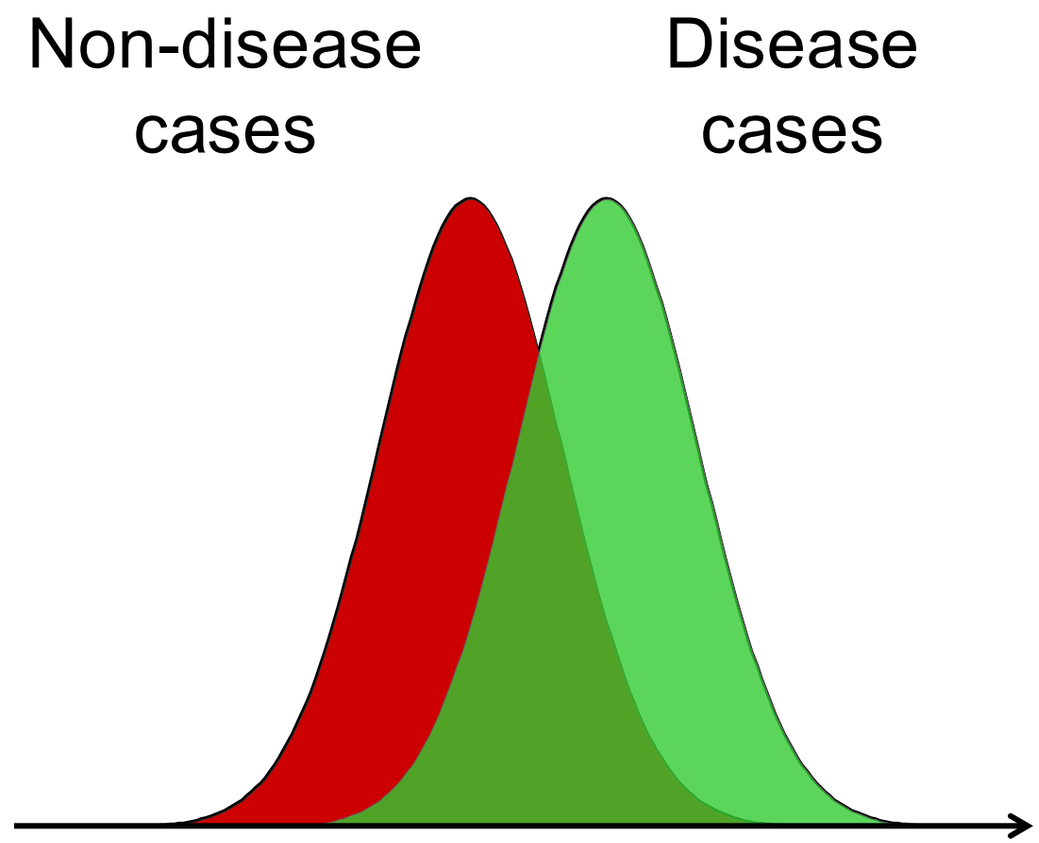
Ideally:



Test result value
or
subjective judgement of likelihood that case has disease

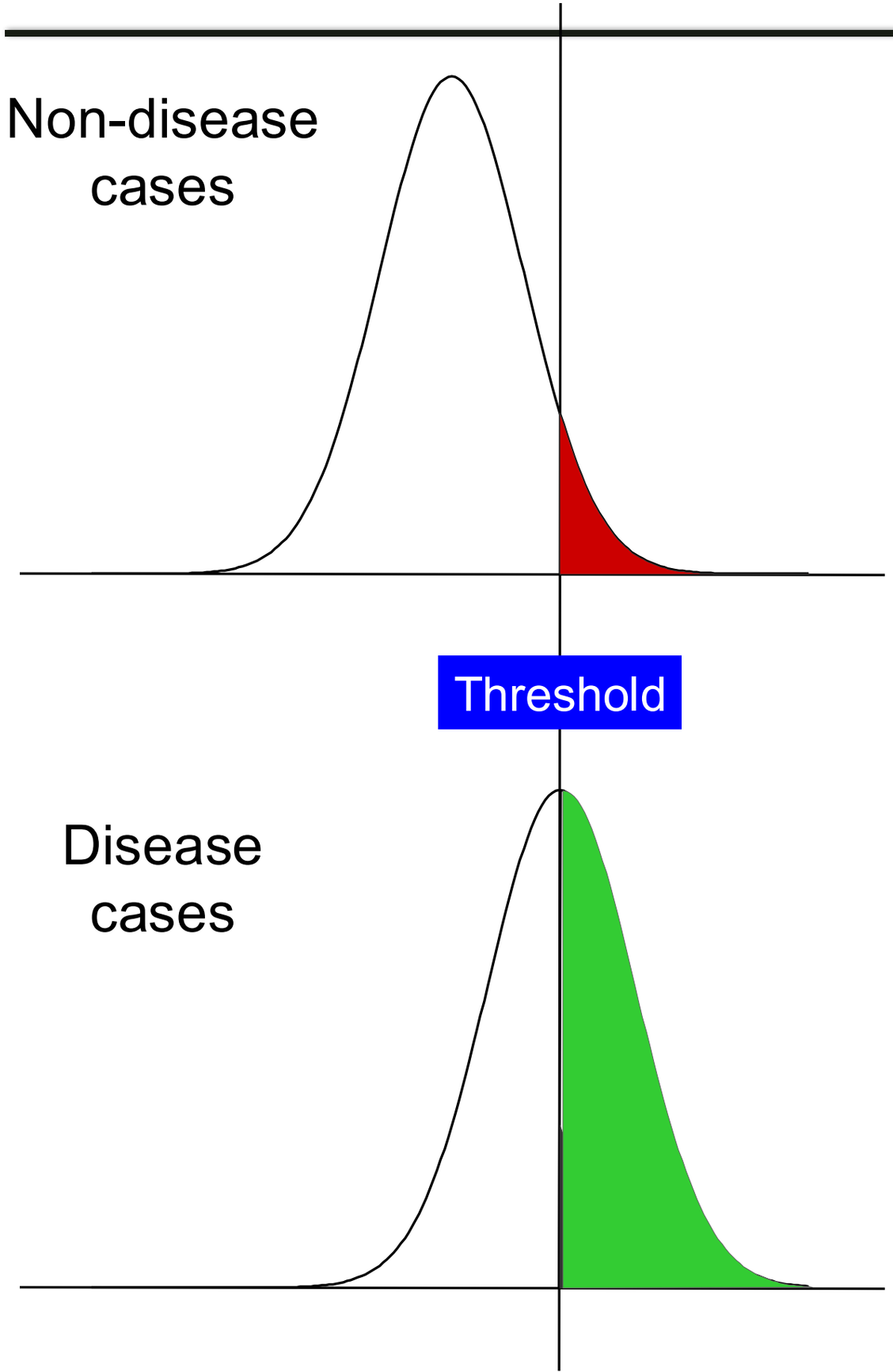
ROC Curve

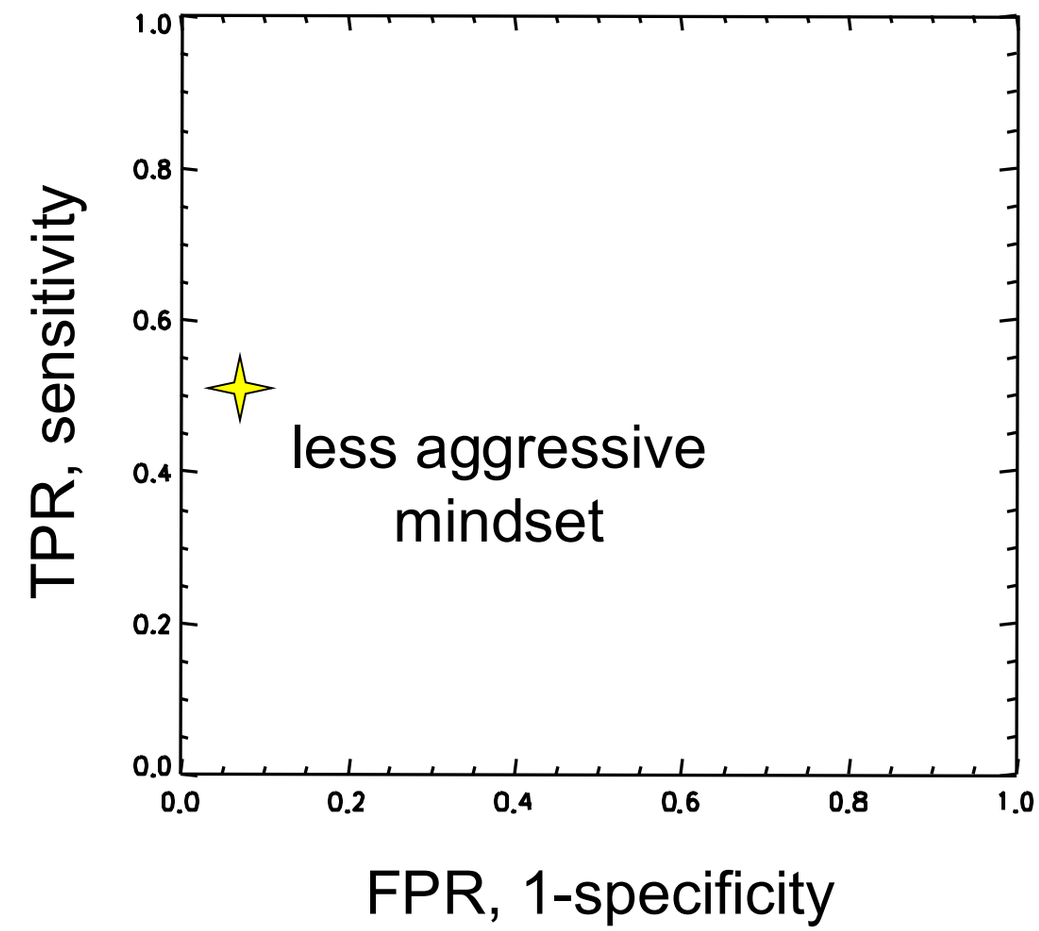
more typically:



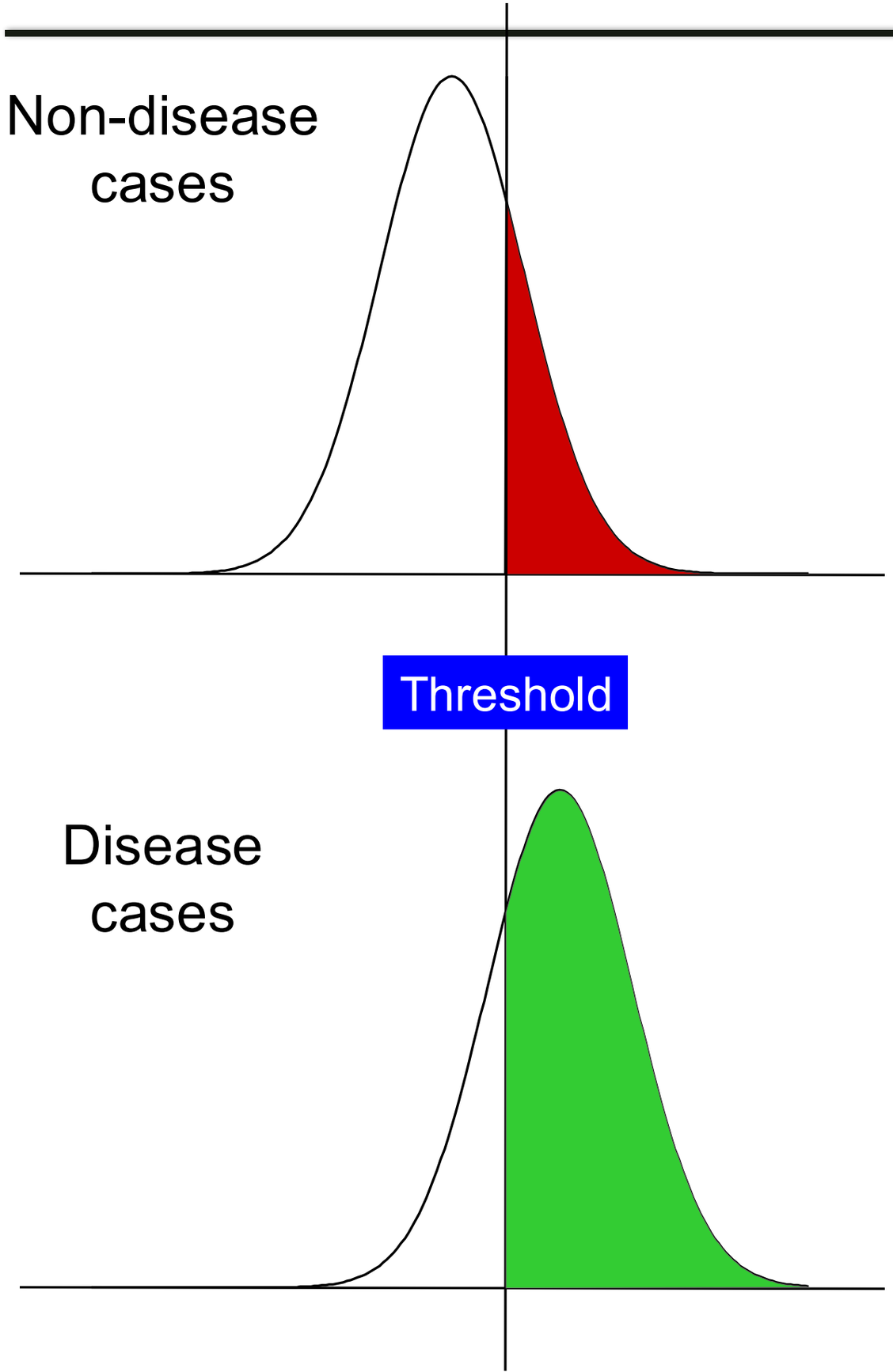
Test result value
or
subjective judgement of likelihood that case has disease

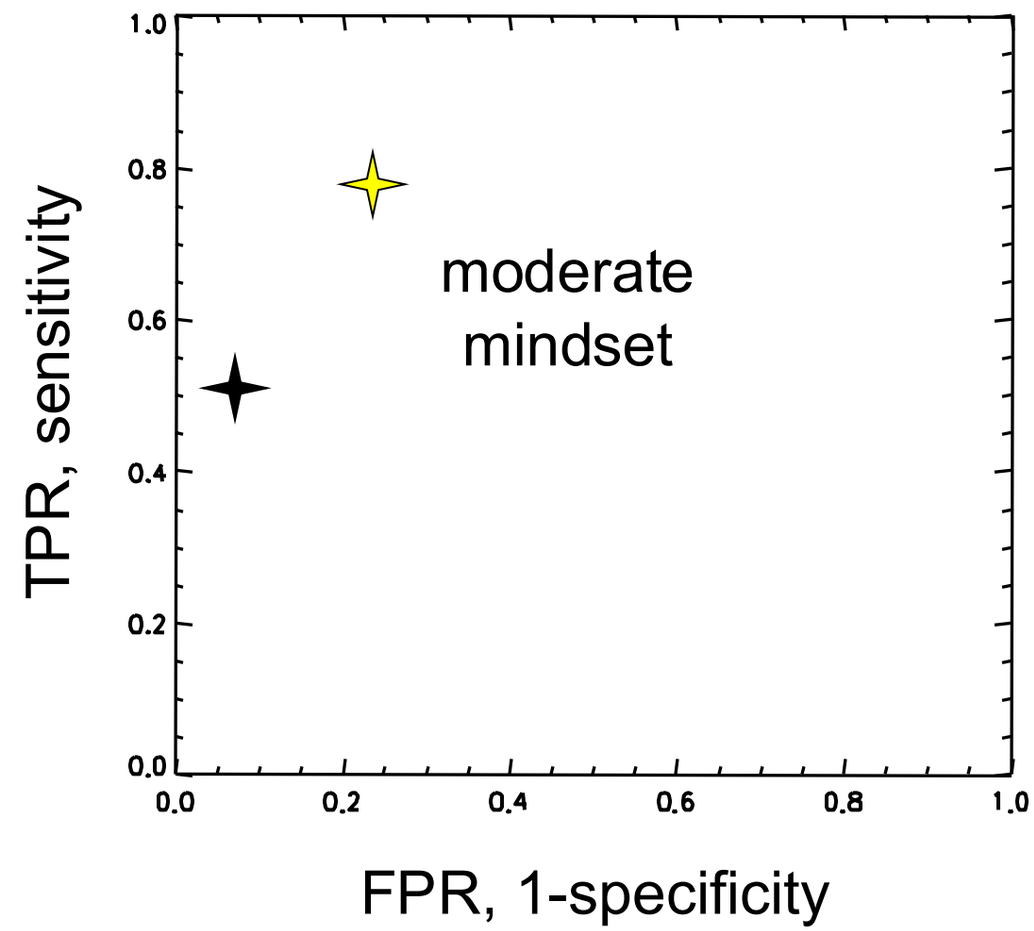
ROC Curve – less aggressive



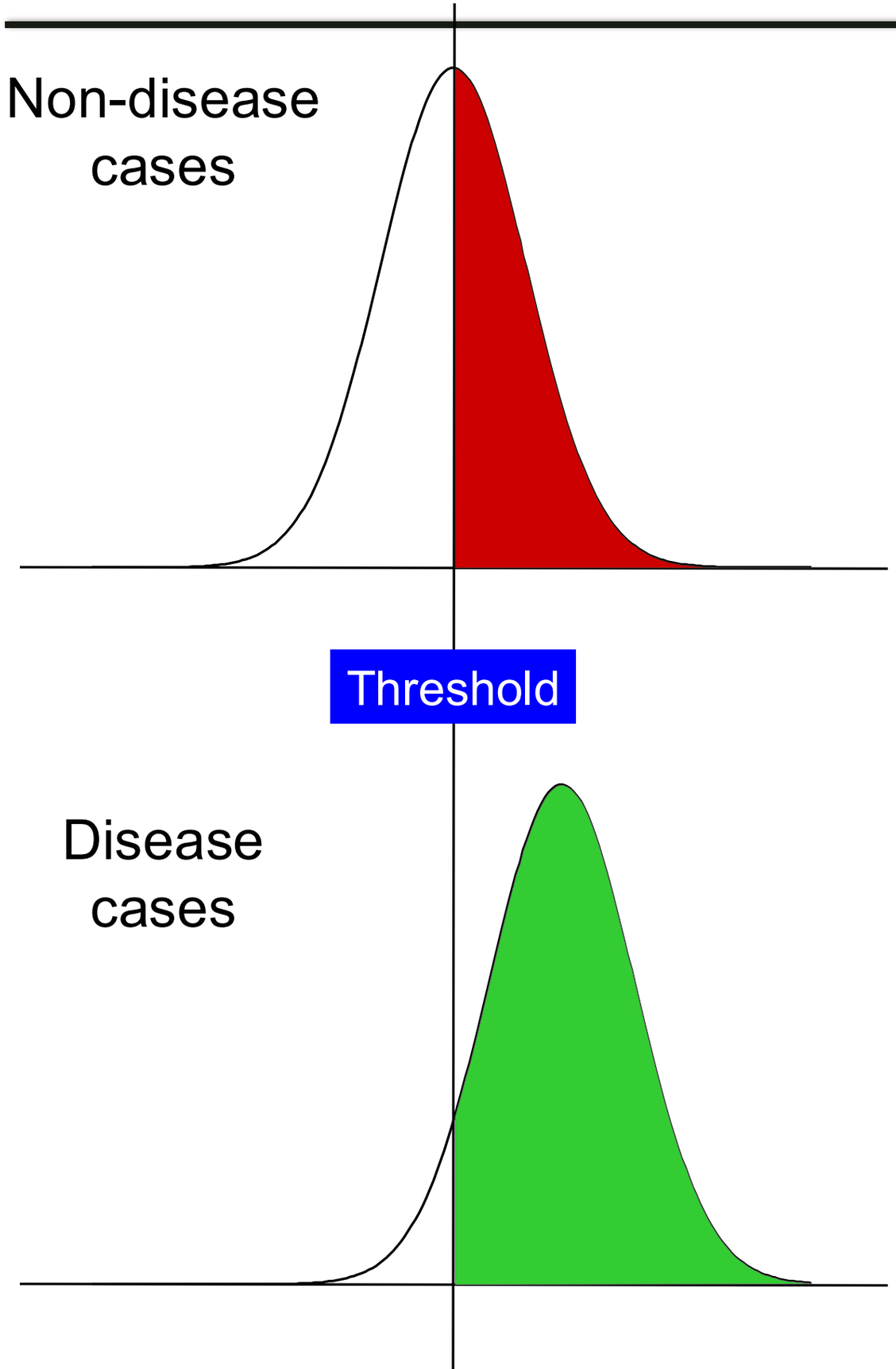


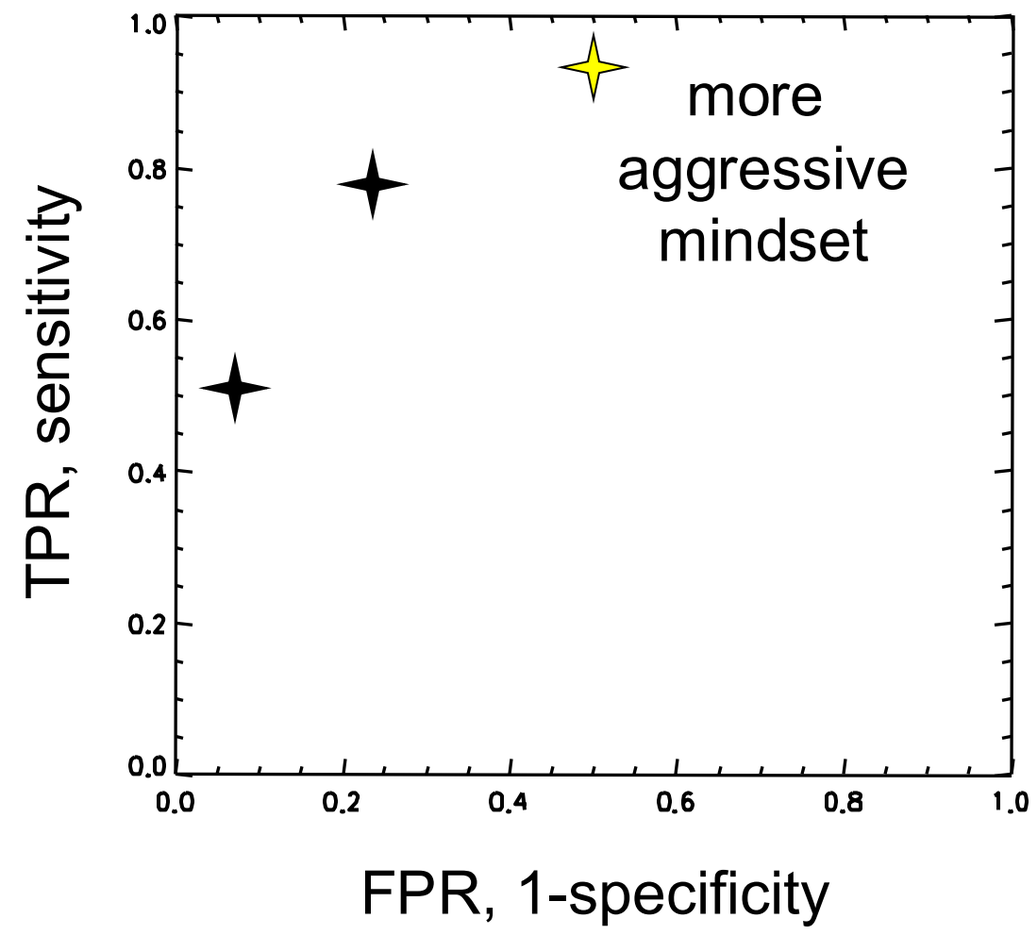
ROC Curve - moderate



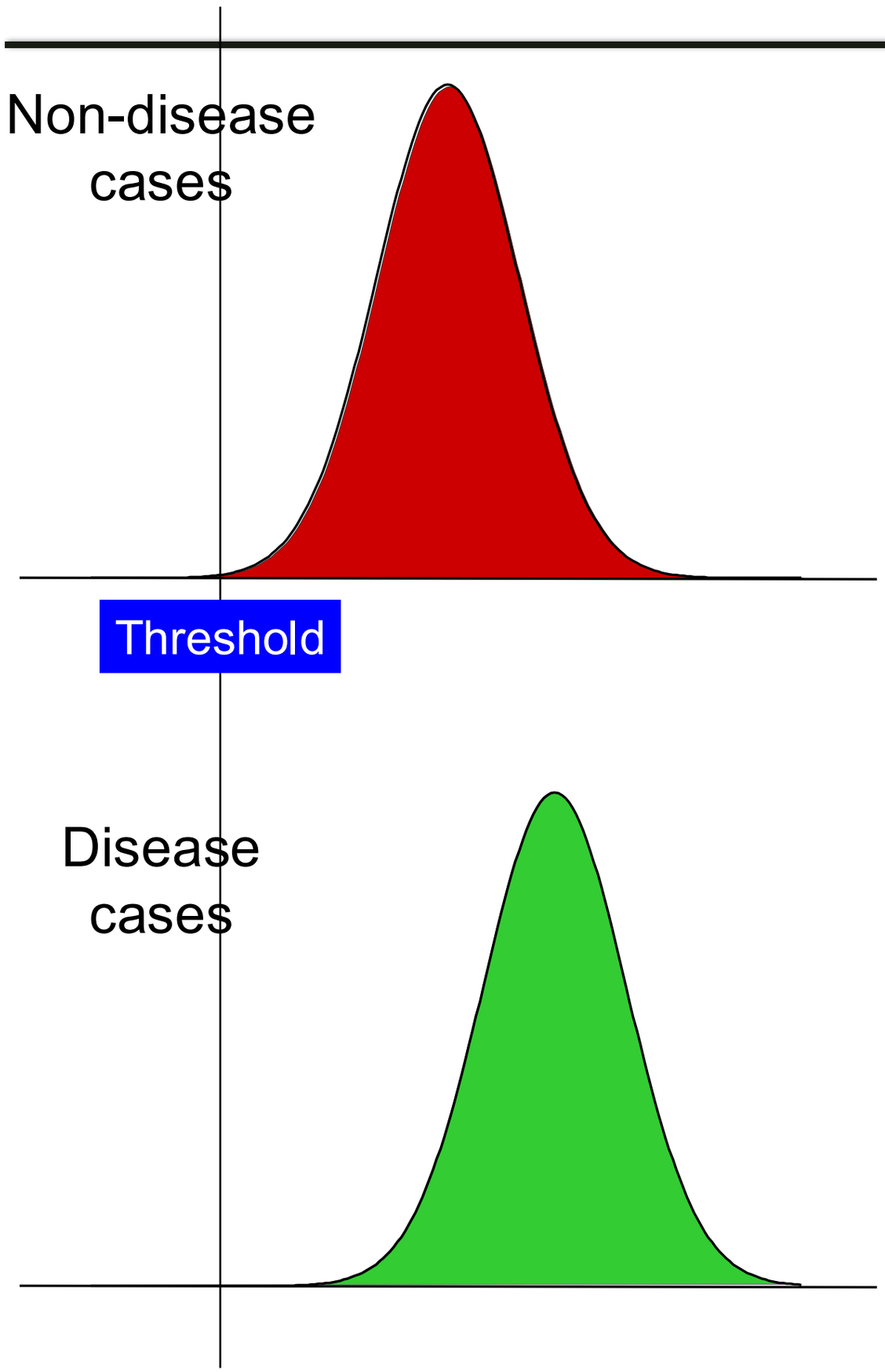


ROC Curve – more aggressive

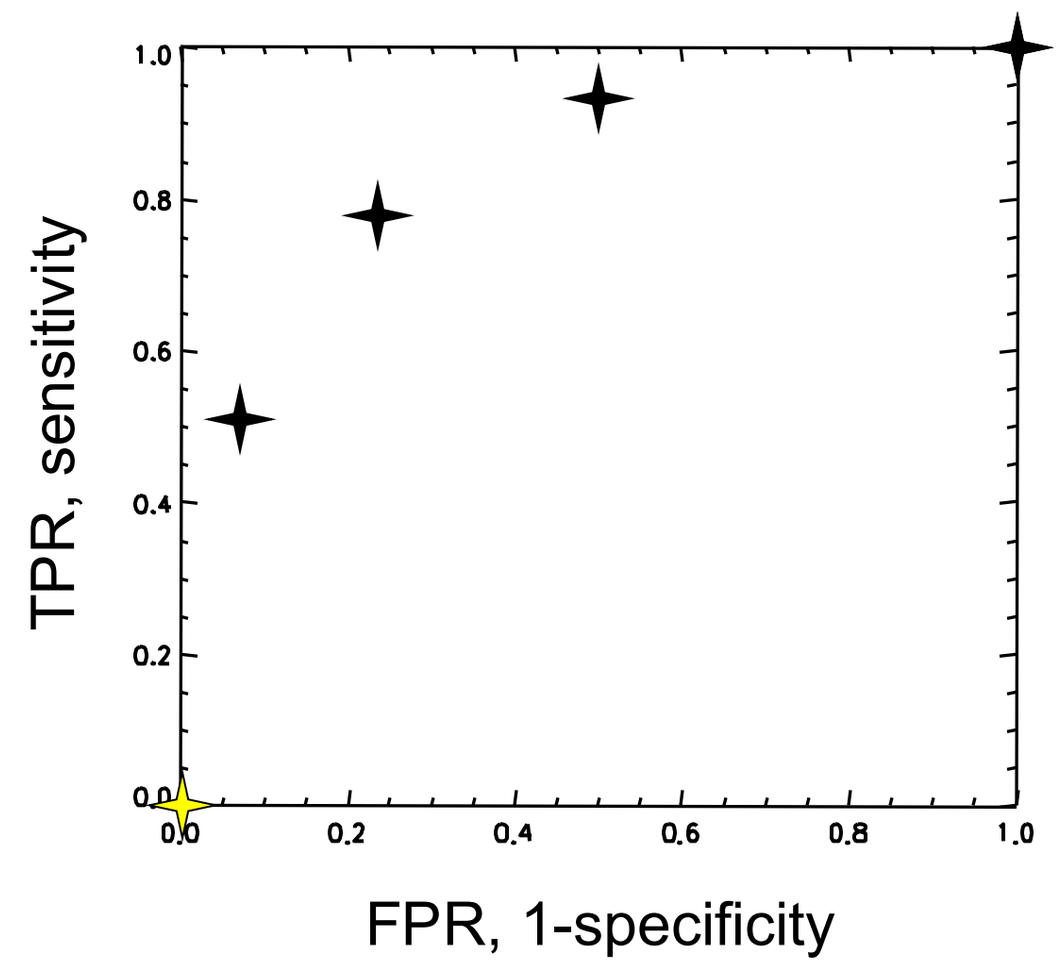




ROC Curve – most aggressive



Entire ROC curve



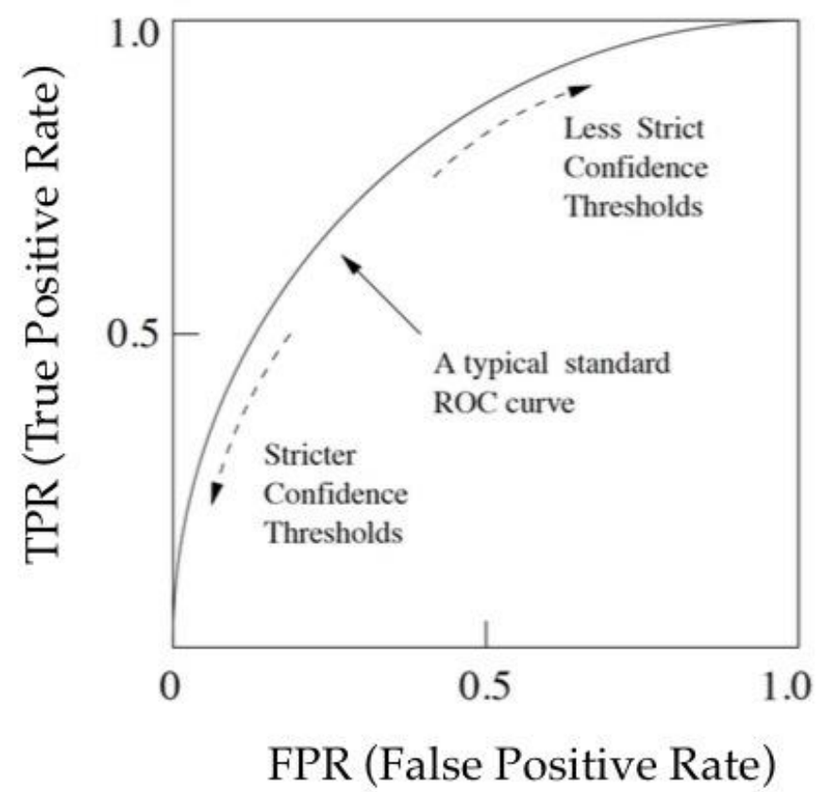
ROC Curve

- Receiver Operating Characteristic (**ROC**) curve:
  - a graphical representation of a classifier's performance across different threshold settings
  - Standard ROC curves conventionally take the False Positive Rate (FPR, 1-Specificity) as the x axis, and the True Positive Rate (TPR, Sensitivity) as the y axis, plot the TPR against the FPR for various decision thresholds.

- Formally, the ROC curve is defined as:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

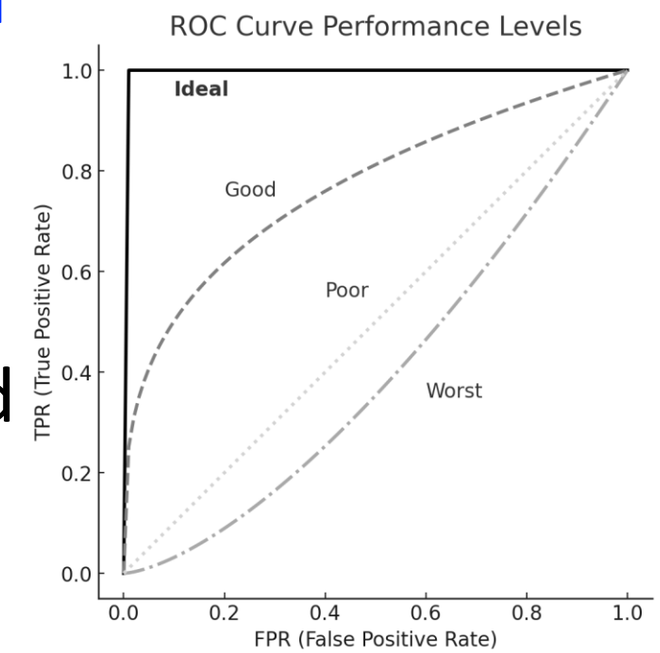
$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$$



(a)

ROC Curve

- Different confidence thresholds correspond to different points, which represent different pairs of TPR and FPR.



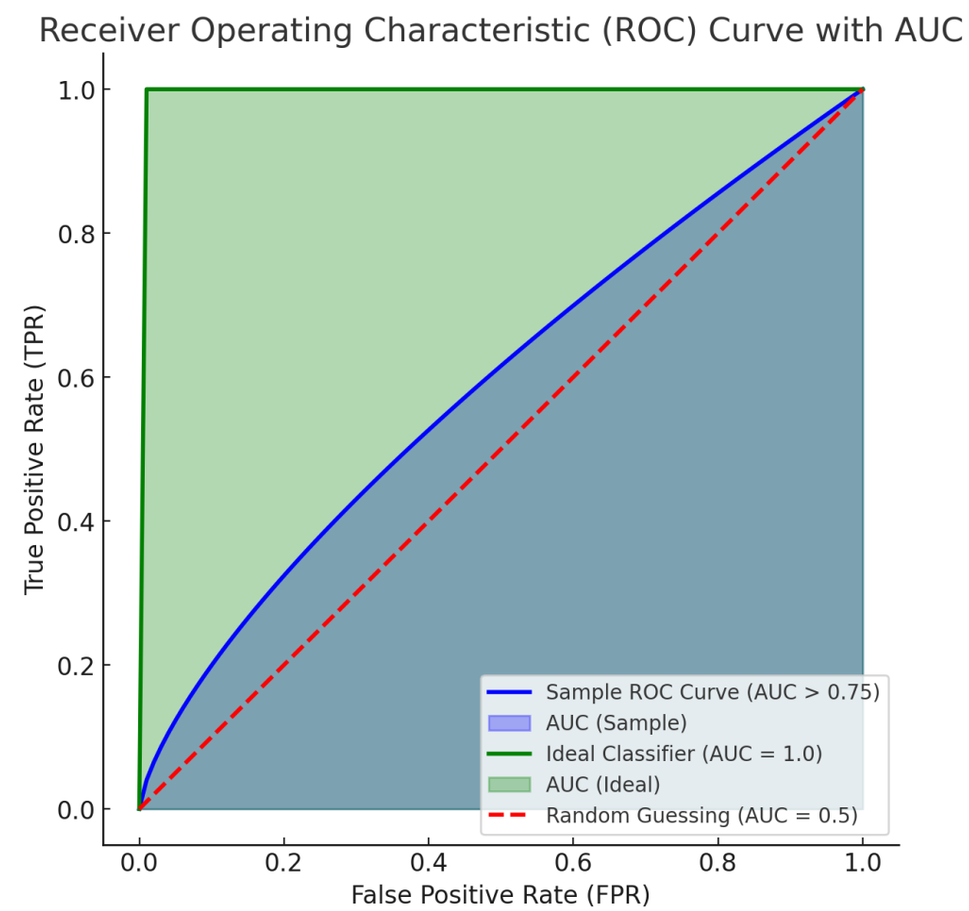
- The ideal system (the “ideal” case): perfect interpretation, and the ideal point is TPR = 1.0 and FPR = 0.
- A higher ROC (the “good” case) curve: greater discrimination capacity
- A lower ROC (the “poor” case) curve: weaker classification capacity
- The worst classification (the “worst” case): has no discrimination between positives and negatives (the “worst” case)

AUC

Conversion of ROC curve to a single measure is necessary sometimes.

One commonly used measure is:

- *Area under the ROC curve* (*AUC*): the area that lies beneath the entire ROC curve.
  - 0.5 < AUC <1.0
  - AUC = 0.5 corresponds to the worst case
  - AUC = 1.0 corresponds to the ideal case
  - Difficult to calculate/estimate, especially when the points are not well spread across the ROC space, heavily depends on the non-interesting part of the ROC curve (both TPR and FPR tend to 1.0).



Example: ROC curves

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import RocCurveDisplay

# Step 1: Generate Synthetic Data
X, y = make_classification(n_samples=1000, n_features=10, n_classes=2, random_state=42)

# Step 2: Split into Training and Testing Sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Step 3: Train a K-Nearest Neighbors (KNN) Classifier
knn = KNeighborsClassifier(n_neighbors=5)  # Using 5 neighbors
knn.fit(X_train, y_train)

# Step 4: Get Probability Scores for the Test Set
y_scores = knn.predict_proba(X_test)[:, 1]  # Get probability for positive class

# Step 5: Display the ROC Curve using RocCurveDisplay
plt.figure(figsize=(8, 6))
RocCurveDisplay.from_predictions(y_test, y_scores)
plt.title("K-Nearest Neighbors (KNN) ROC Curve")
plt.grid()
plt.show()
```

Example Question

- For a two-class classification task, if the classifier correctly achieved the following results under the corresponding thresholds,

	class1					
Threshold	0.40	0.50	0.60	0.70	0.80	0.90
N-O-C	200	200	180	160	120	100
N-O-Tot	500	400	300	240	180	130
N-O-IC	300	200	120	80	60	30
TPR						
FPR						

- Present the ROC curve for class1. Assume we have 200 class 1 cases and 300 class 2 cases.

Precision & Recall

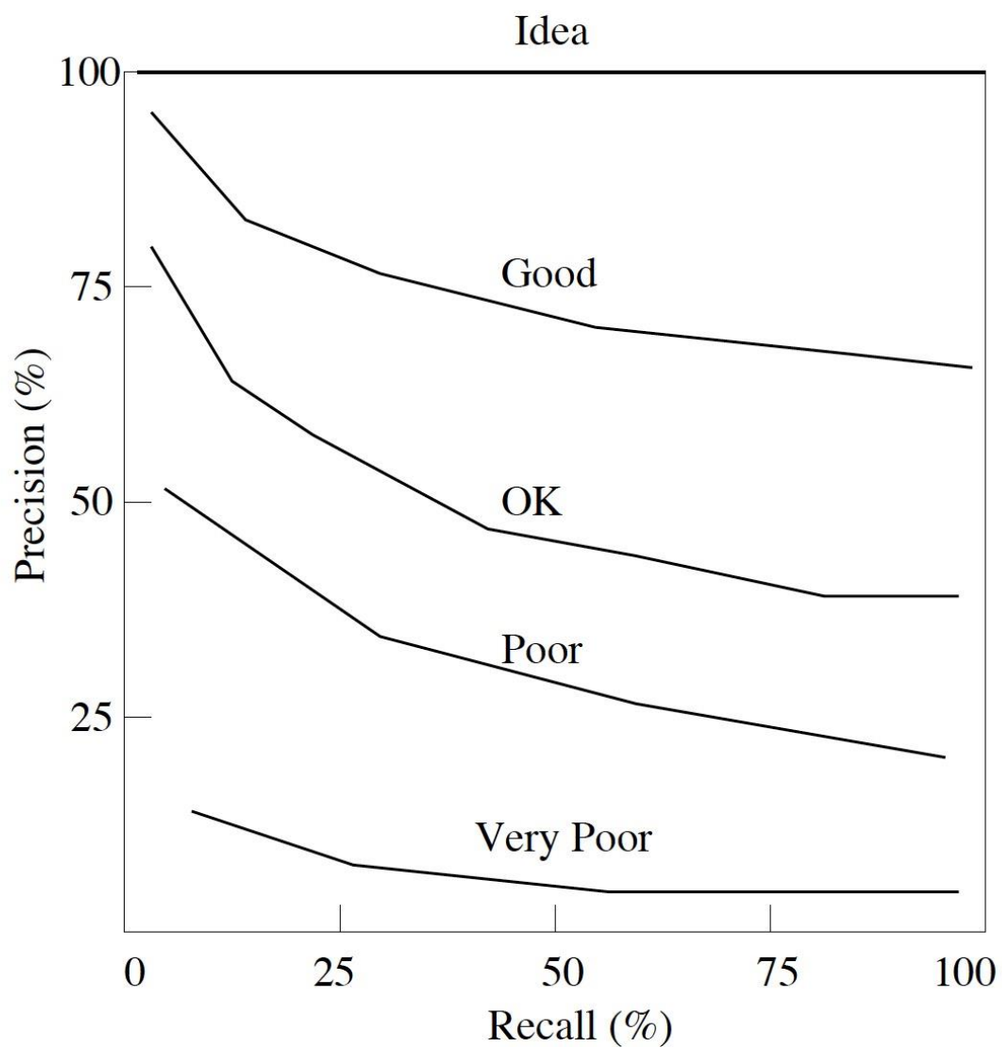
- Precision & Recall: two key metrics used in classification, especially in imbalance classification, e.g., fraud detection, medical diagnosis, and information retrieval.
- Precision measures how many of the predicted positive instances are actually correct.
  - focuses on reducing false positives, making it a crucial metric when the cost of a false positive is high (e.g., mistakenly classifying a benign tumor as malignant)
    - For a single class,
    - $Precision_i = \frac{\sum_{j=1}^{n} N_{true}(i,j)}{\sum_{j=1}^{n} N_{reported}(i,j)} * 100\%$ $\quad Precision = \frac{TP}{TP + FP}$
    - The overall precision for a detection system,
    - $Precision = \frac{\sum_{j=1}^{n} \sum_{i=1}^{m} N_{true}(i,j)}{\sum_{j=1}^{n} \sum_{i=1}^{m} N_{reported}(i,j)} * 100\%$
  - High precision: model predicts a positive instance, it is usually correct
  - Low precision: model frequently misclassifies negative instances as positive

Recall

- Recall: measure how many actual positive instances the model correctly identifies.
  - focus on reducing false negatives, making it important in applications where missing a positive instance has serious consequences (e.g., failing to detect a disease in a patient)
    - For a single class,
      - $Recall_i = \frac{\sum_{j=1}^{n} N_{true}(i,j)}{\sum_{j=1}^{n} N_{known}(i,j)} * 100\%$ $\quad Recall = \frac{TP}{TP + FN}$
    - The overall recall for a detection system,
      - $Recall = \frac{\sum_{j=1}^{n} \sum_{i=1}^{m} N_{true}(i,j)}{\sum_{j=1}^{n} \sum_{i=1}^{m} N_{known}(i,j)} * 100\%$
  - High recall: model captures most of the actual positive cases
  - Low recall: model misses many positive instances
- Recall is identical to detection rate, accuracy, or TPR.
- The range of recall and precision is [0, 1] or [0, 100%].

Precision vs. Recall

- There is typically a trade-off between precision and recall,
  - **Increasing precision** often lowers recall, the model becomes more conservative, avoiding false positives but missing true positives
  - **Increasing recall** often lowers precision, the model tries to capture more positives, including incorrect ones
- some sample precision/recall plots



F1-score

- **F1-score** is the harmonic mean of precision and recall
  - provides a single measure that balances **precision** and **recall**
  - it useful when both false positives and false negatives are important
  - is high only when both precision and recall are high
  - useful in imbalanced datasets, relying only on accuracy can be misleading

$$\textbf{F1 Score} = \frac{2}{\left(\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}\right)}$$

$$\textbf{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Example Question

- Consider a disease screening test where a classifier is used to predict whether individuals have a disease (positive) or not (negative). After testing 100 individuals, the results are as follows:
  - 30 people actually have the disease (positives), and 70 do not (negatives).
  - The classifier identified 25 individuals as having the disease, of whom 20 actually have the disease (TP = 20).
  - The classifier incorrectly identified 5 healthy individuals as having the disease (FP = 5).
  - The classifier correctly identified 65 individuals as not having the disease (TN = 65).
  - The classifier missed 10 cases of the disease (FN = 10).

- Question: What are the precision, recall and F1 score of this classifier?

Performance Metrics Summary

Metric	Formula	Interpretation	When to Use
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness of the model	When classes are balanced
Error Rate	$1 - \text{Accuracy}$	Proportion of incorrect predictions	General performance check
True Positive Rate (TPR) / Sensitivity / Recall	$\frac{TP}{TP+FN}$	Fraction of actual positives correctly identified	Important when missing positives is costly (e.g., medical diagnosis)
True Negative Rate (TNR) / Specificity	$\frac{TN}{TN+FP}$	Fraction of actual negatives correctly identified	Important when false alarms are costly (e.g., fraud detection)
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	False positives relative to actual negatives	Used in ROC curve analysis
False Negative Rate (FNR)	$\frac{FN}{TP+FN}$	False negatives relative to actual positives	Used in ROC curve analysis
Precision (Positive Predictive Value)	$\frac{TP}{TP+FP}$	Fraction of predicted positives that are correct	Important when false positives are costly (e.g., spam detection)
F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall	When balance between precision and recall is needed
AUC (Area Under ROC Curve)	*Computed from ROC Curve*	Overall measure of model performance	Used when different thresholds need evaluation